

Cross-Domain Knowledge Mapping : Biomedical VS AI

Shuvo Chandra Mali.
Department of Computer Science &
Technology
American International University-
Bangladesh
Dhaka, Bangladesh
21-45971-3@student.aiub.edu

Sadia Afrin Moumita
Department of Computer Science &
Technology
American International University-
Bangladesh
Dhaka, Bangladesh
22-49830-3@student.aiub.edu

Rafi Ul Islam Rafi
Department of Computer Science &
Technology
American International University-
Bangladesh
Dhaka, Bangladesh
22-49568-3@student.aiub.edu

Abstract—This paper presents a cross-domain knowledge mapping study that analyzes research abstracts from both biomedical science and artificial intelligence (AI) using text mining, dimensionality reduction, and unsupervised clustering in R. TF-IDF feature extraction, UMAP-based dimensionality reduction, and K-means clustering ($k = 5$) were applied to a merged corpus of diabetics biomedical and AI research papers. The analysis reveals five distinct research clusters, identifies domain-specific vocabulary, and locates knowledge transfer zones where AI methodologies are being applied to biomedical challenges. Clustering metrics including Normalized Mutual Information ($NMI = 0.234$), Adjusted Rand Index ($ARI = 0.263$), and average Silhouette Score (0.333) confirm that the pipeline successfully partitions the two domains and surfaces meaningful cross-domain intersections.

Keywords—text mining, TF-IDF, UMAP, K-means clustering, biomedical research, artificial intelligence, cross-domain knowledge mapping, R

I. INTRODUCTION

The rapid growth of research in both biomedical science and artificial intelligence has created an urgent need to understand how these two fields interact and inform one another. Biomedical research continues to generate large volumes of clinical data spanning patient outcomes, disease mechanisms, and therapeutic interventions while AI research simultaneously advances the algorithmic and computational tools capable of extracting insight from such data. Understanding the structural overlap between these domains can guide interdisciplinary collaboration and accelerate translational research.

This project applies data science techniques to a corpus of research abstracts drawn from both domains, with a focus on diabetes-related biomedical literature and AI research papers. By combining text mining, TF-IDF feature extraction, UMAP dimensionality reduction, and K-means clustering, the study constructs a cross-domain knowledge map that reveals distinct topic clusters, domain-specific vocabulary, and zones of knowledge transfer where AI methods are applied to biomedical problems. All analysis was performed using the R programming language.

II. DATASET AND LIBRARIES

A. Two preprocessed CSV files were used as primary data sources.

First, *Biomedical_Diabetics_DS1.csv* (ds1), contains research abstracts from the biomedical domain, specifically covering diabetes and related clinical topics. The second, *AI_Diabetics_DS2.csv* (ds2), contains abstracts from AI research papers addressing diabetes-related prediction and modelling tasks.

B. Maintaining the Integrity of the Specifications

The analysis relies on the following R libraries: *tm* for text mining corpus management; *ggplot2* for visualization; *umap* for dimensionality reduction; *dplyr* for data transformation; *GGally* for advanced paired visualization; *scales* for axis formatting; and *tidyr* for data reshaping.

III. METHODOLOGY

A. Data Preprocessing

Domain labels were assigned prior to merging: biomedical abstracts (ds1) received label 0 and AI abstracts (ds2) received label 1. The two datasets were combined using *rbind* to produce a single merged dataframe for downstream analysis.

B. Text Cleaning

Raw abstracts were loaded into an *tm* Corpus object via *VectorSource*. A standard cleaning pipeline was then applied: conversion to lowercase, removal of punctuation and numbers, removal of English stopwords, and whitespace stripping. Cleaned text was extracted back into the dataframe using *sapply(corpus, as.character)*. Rows with missing or empty cleaned abstracts were removed, and the resulting clean dataset was saved as *ids_final_dataset_group_11.csv* for reproducibility.

C. Feature Extraction (TF-IDF)

Cleaned abstracts were converted into numerical representations using Term Frequency–Inverse Document Frequency (TF-IDF) weighting. A Document–Term Matrix (DTM) was constructed with *DocumentTermMatrix* and the

weightTfidf weighting function, then converted to a dense matrix for downstream processing. TF-IDF emphasizes terms that are distinctive to specific documents while down-weighting common terms that appear across most abstracts.

D. Dimensionality Reduction (UMAP)

The high-dimensional TF-IDF matrix was reduced to two dimensions using Uniform Manifold Approximation and Projection (UMAP). A fixed random seed (123) was set for reproducibility. The resulting two UMAP dimensions (UMAP1, UMAP2) encode the semantic similarity structure of the corpus in a form suitable for visualization and clustering.

E. Clustering (K-Means)

K-means clustering was applied on the UMAP-reduced coordinates with $k = 5$ clusters, again with seed 123. Each cluster represents a group of research abstracts with similar semantic content. Cluster assignments were stored as factors for visualization. Domain composition within each cluster was calculated as proportional tables, enabling identification of domain-specific and cross-domain clusters.

F. Topic Classification and Keyword Extraction

Clusters were classified into three types based on domain composition thresholds: “Biomedical Specific” if the biomedical proportion exceeded 70%, “AI Specific” if the AI proportion exceeded 70%, and “Shared / Cross-Domain” otherwise. Top keywords per cluster were extracted by computing column-mean TF-IDF scores within each cluster subset and selecting the ten highest-scoring terms.

IV. RESULTS AND ANALYSIS

A. Algorithmic Performance Evaluation

The unsupervised learning pipeline was validated using three standard metrics evaluated against the ground-truth domain labels (Biomedical vs. AI). Table I summarizes the results.

TABLE I. Clustering Performance Metrics

Metric	Value
NMI (Normalized Mutual Information)	0.2340
ARI (Adjusted Rand Index)	0.2631
Avg. Silhouette Score	0.3332

The NMI value of 0.234 indicates a meaningful degree of mutual information between cluster assignments and the true domain labels. The ARI of 0.263 confirms that the clustering is significantly better than random partitioning. An average Silhouette Score of 0.333 reflects moderate-to-strong cohesion within clusters and separation between them, which is a reasonable result given the overlapping vocabulary inherent in cross-domain corpora.

B. Knowledge Mapping and Spatial Distribution

The UMAP scatter plot (Fig. 1) visualizes the cross-domain knowledge map with cluster boundaries drawn as convex hulls. Each point represents one research abstract, positioned such that semantically similar documents appear closer together. Circle markers denote biomedical abstracts (Domain 0) and triangle markers denote AI abstracts (Domain 1).

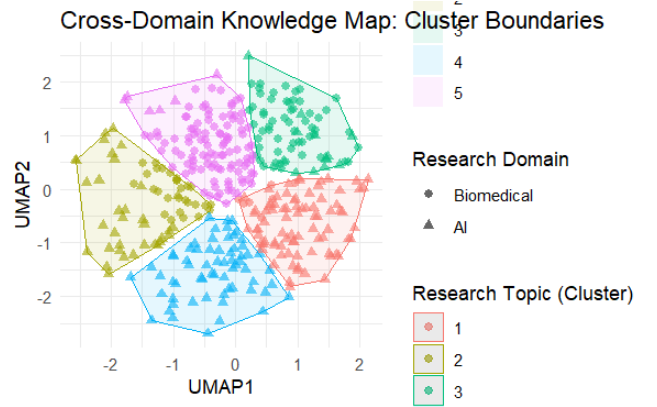


Fig 1: Cross-domain knowledge map with K-means cluster boundaries.

Two key spatial patterns emerge. First, there is clear domain polarization: biomedical and AI abstracts occupy different poles of the UMAP1 axis, confirming that TF-IDF successfully captures the linguistic distinction between the two domains. Second, the histogram of UMAP Dimension 1 values (Fig. 2) shows a bimodal distribution, providing quantitative evidence that the vocabulary of the two domains is mathematically distinct.

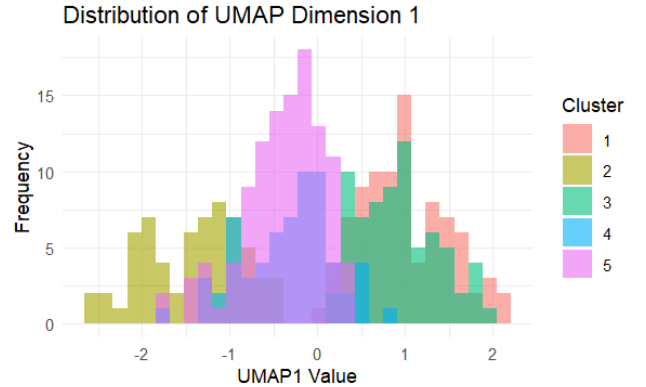


Fig 2: Proportional domain composition across the five extracted clusters.

C. Cross-Domain Knowledge Transfer

The stacked bar chart of domain composition per cluster (Fig. 3) reveals two distinct cluster types. Clusters 1, 3, and 4 exhibit high purity (over 80% of documents belonging to a single domain), representing either deep clinical biomedical research or pure algorithmic AI development. By contrast, Cluster 2 (and to a lesser extent Cluster 5) contains a significant mixture of both domains, forming a “Knowledge

Transfer” zone where AI methodologies are applied to biomedical challenges.

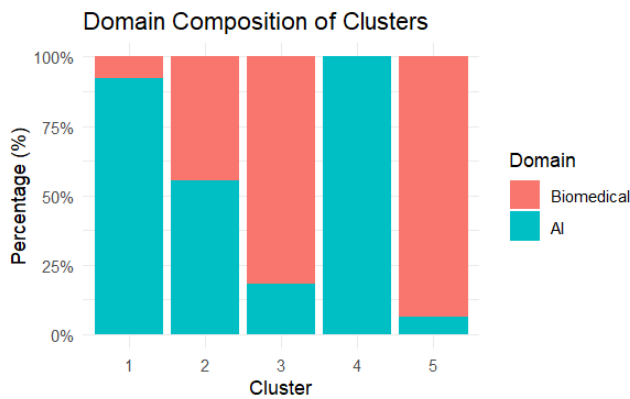


Fig 3: Proportional domain composition across the five extracted clusters.

D. Keyword and Theme Discovery

TF-IDF keyword extraction highlights the semantic content of each cluster. Biomedical-dominant clusters feature terms such as “vascular,” “hyperglycemia,” “insulin,” and “complications,” reflecting clinical and physiological terminology. AI-dominant and cross-domain clusters surface terms including “forecasting,” “benchmark,” “machine-learning,” and “predictive-models.” The co-occurrence of these term sets in the transfer-zone clusters indicates that clinical data is being actively integrated into computational frameworks to improve healthcare outcomes.

E. Comparative Stability Analysis

The box plot of UMAP Dimension 2 variance by cluster (Fig. 4) and the violin plot of domain density distributions (Fig. 5) together reveal an important asymmetry. AI research abstracts (Domain 1) exhibit a more concentrated density, consistent with the standardized technical vocabulary that characterizes the field. Biomedical research abstracts (Domain 0), on the other hand, show considerably wider spread, reflecting the linguistic breadth of a domain that spans molecular biology, clinical trials, epidemiology, and patient care. This variance analysis reinforces the view that biomedical literature is structurally more diverse than AI literature at the vocabulary level.

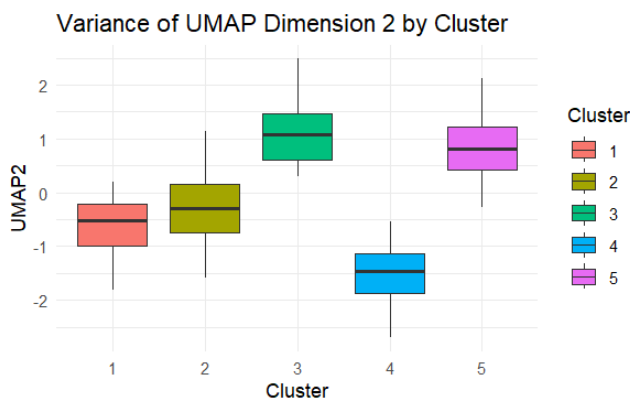


Fig 4: Internal quantile variance of UMAP Dimension 2 by cluster.

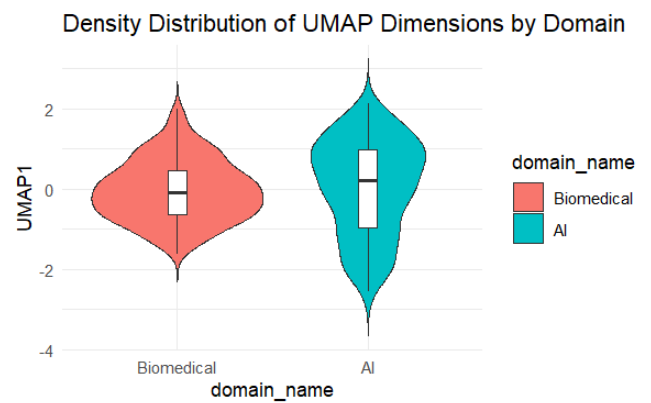


Fig 5: Smooth violin density curves of UMAP dimensions by domain.

Scatter Matrix of UMAP Dimensions

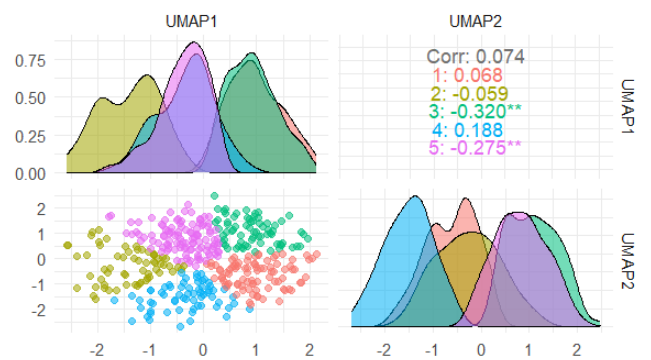


Fig 6: Scatter plot matrix displaying pairwise UMAP coordinate interactions.

V. CONCLUSION

This study demonstrates that an unsupervised text mining pipeline combining TF-IDF, UMAP, and K-means clustering can effectively map knowledge structures across biomedical and AI research domains. Five meaningful clusters were identified, some representing domain-specific topics and others constituting cross-domain transfer zones where AI is being applied to biomedical problems. Quantitative validation metrics (NMI, ARI, Silhouette Score) confirm the statistical validity of the partitioning, and keyword extraction provides interpretable labels for each cluster.

The findings highlight the growing convergence of AI and biomedical research, particularly in diabetes-related predictive modelling and clinical decision support. Future work could extend this analysis with larger corpora, apply dynamic topic modelling to track the evolution of cross-domain themes over time, and incorporate citation network analysis to further characterize knowledge transfer pathways.

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